ELA: Recursive Linguistic Elements & the Context–Pattern Duality

An AI-Assisted Framework for Authentic English Comprehension — Where does language end and pattern begin?

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github.com/welsol21/FYP_LLM

Problem & Motivation

Language-learning platforms use **pre-simplified materials**. Authentic English — films, podcasts, news — is inaccessible without structural guidance. Simplification destroys the input.

- Can NLP + AI *explain* authentic text without altering it?
- What recursive data model stays human-interpretable?
- Can bilingual annotation reduce cognitive load?

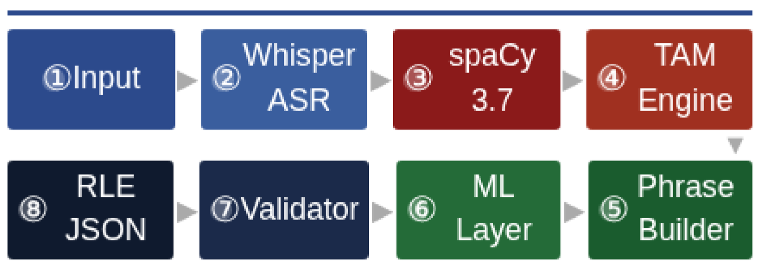
The RLE Data Model

Sentence → Phrase → Word — nested JSON validated against a dual-schema contract.

```
Sentence → Phrase
(NP/VP/PP/ADJP) → Word
                    └──
translation · CEFR · POS · IPA ·
TAM · T5 note
```

Language-agnostic. English → Russian via M2M100 (offline, on-device).

Analysis Pipeline



PWA — Android-installable, fully client-side, no cloud.

Research Journey: What We Tried

CEFR
- ~~DeBERTa~~ ~~underperformed~~ → ~~BERT~~ ~~same~~ →
- XGBoost ✓ 78.3%

Notes
- ~~GPT-4o~~ ~~cost/cloud~~ → ~~T5 raw~~ ~~no generalise~~ →
- T5+Templates ✓ ROUGE-L 0.81

ASR
- Whisper ✓ WER 8.2%

Translation
- ~~GPT-4o~~ ~~opt-in~~ → M2M100 ✓ offline

The New Physics of Context–Pattern Interaction

> **"In open systems, context forms patterns.
> In closed systems, patterns form context."**

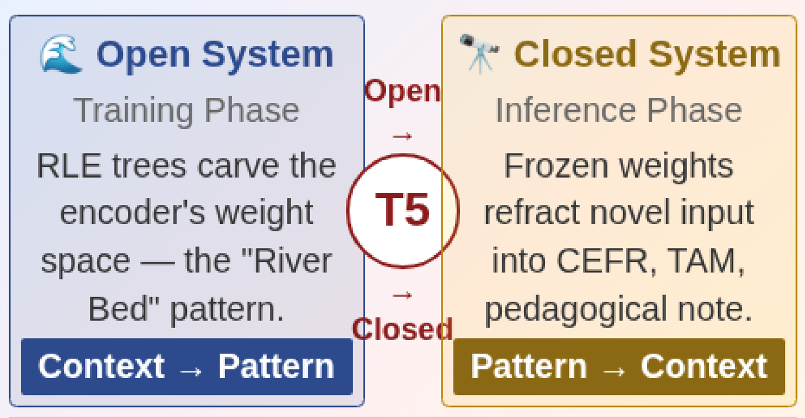
— Discovered empirically during T5 fine-tuning

T5 on raw notation **failed to generalise**. Structural placeholders were the breakthrough:

```
❌ "She gave him a book" → [VP gave][NP him]
✅ [SUBJ][V-PAST][IOBJ][DET][NOUN] → [TAM: past-simple]
```

*Abstract grammatical patterns — invariant to lexical substitution — impose structure on novel input. This is **closed-system behaviour**.*

The Duality Loop Diagram



Phase	System	Mechanism
Training	Open	RLE trees shape encoder weights
Inference	Closed	Frozen weights interpret novel input

Training Data & Corpus Scale

Corpus	Tokens
OANC	30M+ tokens
UD GUM	5.2M tokens
Gutenberg	1M+ tokens
MASC	~500k tokens

27k→685 unique · **6 CEFR** A1→C2

Live Demo
www.el-a.uk
Try it in your browser

Performance Metrics

78.3%	0.81	8.2%
CEFR Accuracy XGBoost ensemble	ROUGE-L T5 fine-tuned	ASR WER Whisper base

Annotation Backend Comparison

Approach	Quality	Cost	Offline
GPT-4o	High	$$$	✗
Rules only	Medium	Free	✓
T5 fine-tuned	**High**	**Free**	✓
XGBoost CEFR	78.3%	Free	✓

System Validation

- **O1:** spaCy + TAM → schema-valid RLE ✓
- **O2:** CEFR 78.3%; T5 ROUGE-L 0.81 ✓
- **O3:** Visualizer + JSON/CSV export ✓
- **O4:** Android PWA, fully client-side ✓

Conclusions

- Explainable AI annotation makes authentic English **pedagogically accessible without simplification**.
- The **Context–Pattern Duality** generalises to any open→closed learning system — training carves patterns, inference imposes them.
- Placeholder-driven T5 is **lexically invariant** — generalises to vocabulary unseen during training.

Future Directions

- **Tauri 2 Desktop App** — native .deb/Windows with bundled ML models.
- **Translation Quality Gates** — ChrF scorer, M2M100 review queue.
- **M2M100 LoRA Fine-Tuning** — domain-specific EN→RU refinement.
- **Real-time Streaming ASR** — Whisper streaming for live practice.

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